RETAIL CREDIT SCORING

TEAM NINE
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RETAIL CREDIT SCORING

Overview

The Commercial Banking Corporation's department of Revolving Lines of Credit seeks to enhance its credit evaluation process and optimize profit by either maintaining its current 75% acceptance rate while lowering the 3.23% default rate, or maintaining the current default rate while maximizing the acceptance rate. Our team was tasked with developing a comprehensive credit scoring solution that creates a predictive scorecard, establishes risk-based score distributions, implements reject inference, and identifies optimal cutoff strategies to balance acceptance rates with profitability.

We developed a scorecard that demonstrates strong discriminatory power while providing actionable insights for credit decision-making. Based on the results, we recommend that the Bank adopt our scorecard and utilize our cutoff ranges so that anyone with a score above 483 is automatically approved, anyone with a score below 428 is automatically declined, and anyone between these scores is referred to manual adjudication. The utilization of such cutoffs will maintain our 3.23% default rate while maximizing profit at approximately $8 million.

Methodology & Analysis

This section discusses the dataset provided, how our team prepared it for modeling, and the analyses our team performed to evaluate retail credit applications.

Data Used

Our team received an accepted customer dataset that contained 3,000 observations and 24 variables. This dataset was balanced so that there are 1,500 "good" accepted applicants and 1,500 "bad" accepted applicants. Bad, in this dataset, was defined as an applicant being 90 days past due at least once. Anyone that is not considered bad is considered good. We used the weight variable provided in the data, which reflected the current default rate of 3.23%. This weight variable assigned good applicants a weight of 30, while bad applicants were assigned a weight of one.

Our team also received a rejected customer dataset that contained 1,500 observations and 22 variables. This dataset contained information on the rejected applicants and was used to conduct reject inference.

Data Cleaning

We normalized the column names in the accepted and rejected customer datasets. We also split the accepted data into a train and validation dataset using a 70% and 30% split, as suggested by the Bank.

Variable Grouping and Selection

Our team then performed multivariate binning of the first 22 columns (excluding the target and frequency variables). The binning process used an automatic binning procedure designed to select variables based on information value (IV) criteria. Our team chose to select binned variables that were within an IV range between 0.1 to one.

Scorecard Creation

Our team fit the scorecard using logistic regression and points to double the odds (PDO) scaling. The Bank assigns a score of 500 to applicants with an odds ratio of 20:1 and doubling the odds is associated with a change of 50 points in the scorecard. These qualities were used to adjust the PDO scaling while fitting our scorecard. Our team evaluated this scorecard on both the training and validation dataset using the receiver operating characteristic (ROC) curve and the Kolmogrov-Smirnov (KS) test.

Creation of Decile Plot by Default Rate

We re-read the accepted customers data in and created the same splits for the train and validation datasets that were formed before variable grouping and selection. This allowed our team to use the previously fitted scorecard on the original dataset in order to automatically apply the learned bins from before and calculate the scores for the observations. By doing so, we were able to form deciles and split the counts of the good and bad applicants amongst said deciles. We also computed the probability of default (PD) and from this, and the previously defined deciles, we were able to obtain the PD per decile.

Reject Inference

Due to our team having access to the rejected customer data, we performed reject inference. We used the KS statistic to determine a cutoff value of 3% on the rejected customer dataset, indicating that those with a PD greater than 3% would be classified as bad while those equal to 3% or less would be considered good. To account for selection bias,we concatenated the full rejected applicants dataset with the 3,000 accepted applicants. This combined population was then split into training and test datasets at a 70% and 30% ratio respectively. We applied new weights of 29.96 and 10.32 to the good and rejected applicants, respectively, to maintain the 3.23% default rate while simultaneously preserving the weight of one for bad applicants. Finally, we recreated the scorecard using the same steps previously done on the train dataset and then used the test dataset to evaluate the scorecard.

We continued to use the test dataset of the combined data to create a new PD by score decile plot, a default rate versus acceptance rate across score cutoffs line graph, and a profit versus acceptance rate across score cutoffs line graph. When creating the final graph regarding profit, we used pre-determined expected revenue and cost values from the Bank. This indicated that the expected revenue of accepted good customers is about $2,000 while the expected cost of accepted bad customers is $52,000.

Results

This section presents findings from applying the scorecard model to predict default risk among applicants.

Variable Selection

The multivariate binning process identified several key variables for the final scorecard. These were selected based on the IV thresholds detailed in the methodology section. The IV represents each variable's ability to distinguish good from bad applicants through its binned structure. The Gini coefficients of each variable are also reported to reflect its overall discriminatory performance. **Table 1** displays the primary predictors selected for the final scorecard.

Table 1: Variable Importance Metrics of Selected Variables

Variable	Information Value	Gini Coefficient
Age	0.853	0.488
Time at Job	0.462	0.353
Income	0.446	0.364
Household Size	0.302	0.278
Credit Card Count	0.300	0.234
EC Card Indicator	0.240	0.204
Type of Credit Product	0.130	0.196
Professional	0.113	0.128
Time at Address	0.110	0.182

The variable selection results indicate that age, time at job, and income serve as the primary anchors of the scorecard, having the highest Information Value and Gini coefficients. These variables demonstrate the strongest discriminatory and rank-ordering performance among the variables and account for the majority of the model's ability to differentiate between good and bad applicants.

Scorecard Model Evaluation

The scorecard demonstrated strong discrimination between good and bad applicants, achieving an AUC of 0.770 and a K-S Statistic of 0.432 on the test dataset. This discrimination is illustrated in **Figure 1**.

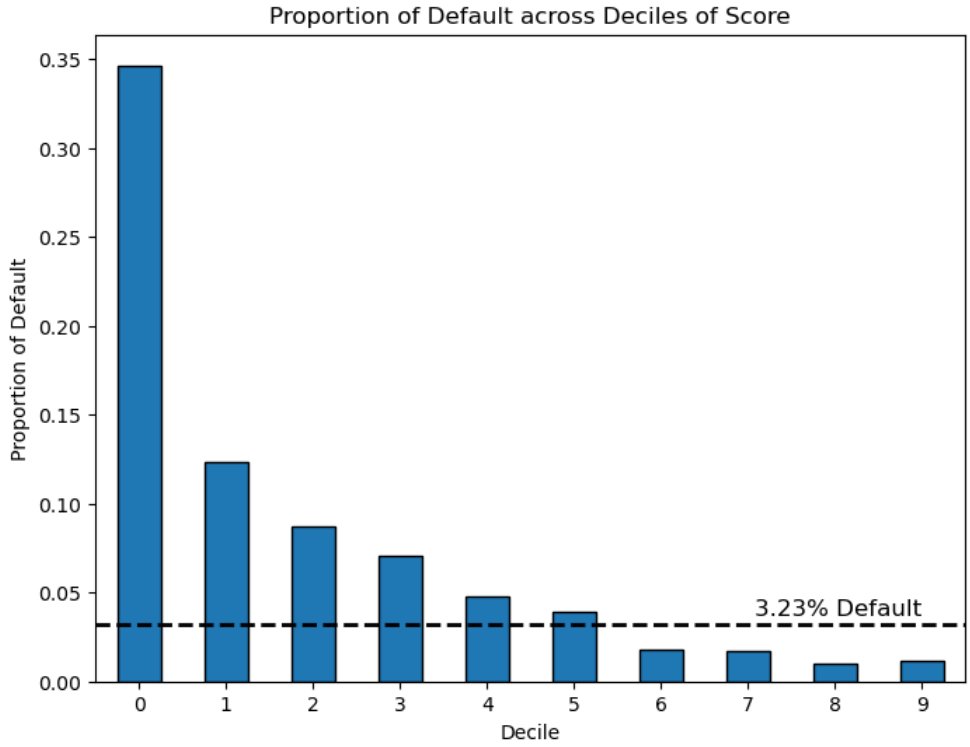

Figure 1: Proportion of Default across Score Deciles

As shown in **Figure 1**, the scorecard demonstrates strong rank-ordering power, with the default rate typically decreasing as scorecard deciles increase. This supports the desired monotonic relationship of the scorecard, with the highest default rate concentrated in the lower-score deciles. Minor fluctuations are observed in higher-score deciles, such as Deciles 8 and 9. This isolated break in monotonicity may be driven by a small number of observed defaults in low-risk deciles, where individual default events can disproportionately affect the estimated default rate.

Evaluating Score Cutoff

We evaluated alternative score cutoffs under three strategic objectives: maintaining the Bank's historical acceptance rate, maintaining the historical default rate, and maximizing expected profitability. **Figure 2** presents the expected cumulative profit relative to the acceptance rate, assuming an expected revenue of $2,000 per accepted good customer and an expected loss of $52,000 per accepted bad customer.

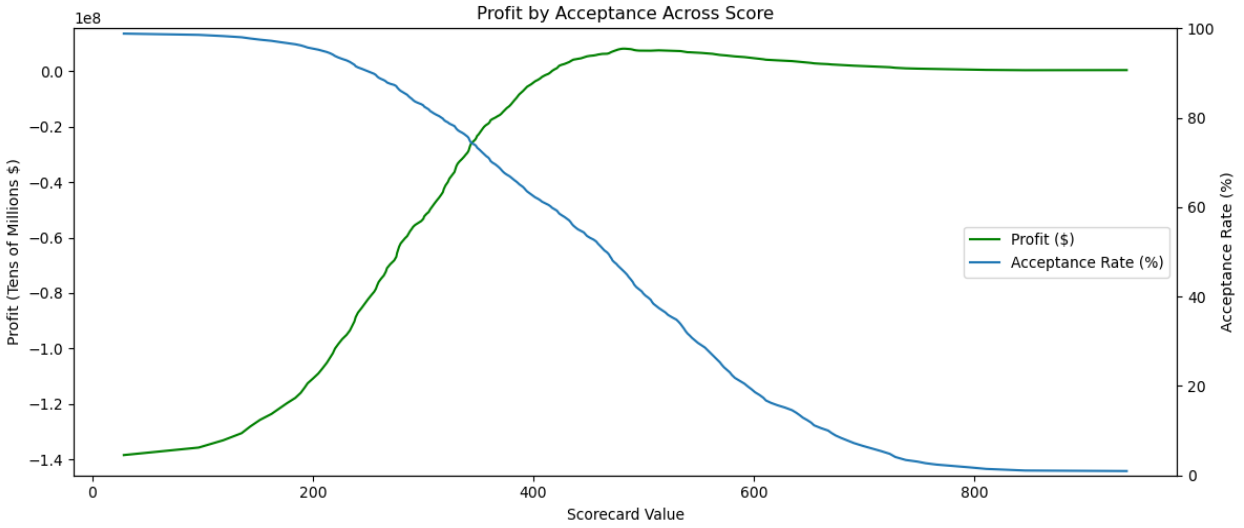

Figure 2: Profit by Acceptance Rate across Scores

Figure 2 demonstrates that as acceptance rates increase, expected profit initially rises due to higher loan volume. However, beyond a certain point, profitability sharply declines as the marginal cost of additional defaults outweighs revenue from newly accepted applicants. This reflects the highly asymmetric cost structure, in which a single default offsets the revenue generated by 26 good customers. **Table 2** compares the outcomes across the three candidate strategies.

Table 2: Strategic Score Cutoff Comparison

Strategy	Score Cutoff	Acceptance Rate	Default Rate	Total Expected Profit (Millions)
75% Acceptance Rate	342	75%	7.42%	-27.94
3.23% Default Rate	428	57%	3.23%	2.57
Maximize Profit	483	45%	1.92%	8.08

Based on **Table 2**, maintaining the Bank's historical default rate of 3.23% requires a more conservative 57% acceptance rate, which yields a limited profit of $2.57 million. Maintaining the historical acceptance rate of 75% would have a significant negative impact on profit, resulting in an expected loss of $27.94 million due to the associated 7.42% default rate. In contrast, the profit-maximizing strategy identifies a 'sweet spot' at a 483 score cutoff, resulting in a 45% acceptance rate. Although this targets a lower volume of applicants, it achieves the highest profit of $8.08 million by effectively controlling the default rate at 1.92%, demonstrating that a high volume of acceptances is unsustainable given current risk levels and the asymmetric costs of good and bad applicants.

Recommendations

Based on our analysis, we recommend the following actions to the Commercial Banking Corporation's department of Revolving Lines of Credit:

Scorecard Implementation

Implementing the scorecard our team created to evaluate all retail credit applications will maintain the existing default rate of 3.23% while simultaneously maximizing profit. The scorecard we created performs very well at distinguishing between the Bank's good and bad customers, as seen by the AUC of 0.770, and the implementation will expedite the acceptance or rejection of applications. The Bank has the possibility to obtain a profit of $8,078,596 by utilizing the scorecard, thus allowing for possible expansion of the Bank's services.

Utilization of Cutoff Ranges

The Bank should begin applying the cutoff ranges that our team created such that scores:

- Above 483 will automatically be accepted (maximizes profit)
- Below 428 will automatically be declined (maintains 3.23% default rate)
- Between 428 and 483 will be subject to manual review

The utilization of such cutoff ranges will allow the Bank to improve efficiency for the applicant process and reduce costs, as the automation of such processes will greatly reduce the amount of manual reviews that are necessary. Applying these cutoff ranges will maintain consistency across the Bank while maximizing profitability, reducing risk, and maintaining the default rate.

Conclusion

Our analysis has successfully delivered a robust credit scoring solution that meets all of the Bank's objectives while providing clear opportunities for enhanced profitability. The developed scorecard demonstrates strong discriminatory power with an AUC of 0.770 and effectively distinguishes between good and bad applicants, with default rates decreasing monotonically from over 10% in the lowest score decile to less than 1% in the highest.

The scorecard's performance is driven by key variables, including age, time at job, and income, providing a comprehensive risk assessment framework. The cutoff ranges created will drive profit maximization and maintain the Bank's 3.23% default rate. The Bank's use of such cutoff ranges will result in reduced costs through automation, reduced risk, and more consistent behavior across the Bank. Implementation of this scorecard will enable more informed, consistent, and profitable credit decisions while maintaining regulatory compliance, ultimately strengthening the Bank's competitive position in the retail credit market.